Mathematical and Computational Methods

Lecture 5: Matrices

Lecture 4 ended on a promise: the lines and planes we wrote down are the solution sets of *linear systems*, and asking where they meet is asking to solve one. This lecture builds the tool. A *matrix* is a rectangular array of numbers that plays two roles at once: it *stores* the coefficients of a linear system, and it *acts* on vectors as a linear transformation—a rotation, a reflection, a stretch. We develop the algebra of matrices, then the two numbers that govern a square matrix—its *determinant* and its *inverse*—and put them to work solving systems and transforming the plane.

Where this fits. The determinant met informally in Lecture 4—the mnemonic for the cross product, and the triple product that measured volume—is defined here in its own right, and the vectors of Lecture 4 are the objects a matrix multiplies and the columns a system is built from. For physics students this is the only place in the course matrices appear, so the unit is self-contained. Looking ahead, Lecture 6 needs all of it: eigenvalues come from a determinant equation, $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$; eigenvectors from solving a system by the elimination developed here; and diagonalisation from an inverse.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Perform matrix algebra and check multiplication compatibility; use the transpose and trace.
- Compute determinants and inverses of 2×2 and 3×3 matrices by hand.
- Solve linear systems by the inverse method and by Gaussian elimination.
- Interpret solution sets geometrically and apply transformation matrices.

Why matrices? A system to return to

Consider the three equations in three unknowns

$$\begin{aligned} 2x + y - z &= 8, \\ -3x - y + 2z &= -11, \\ -2x + y + 2z &= -3. \end{aligned}$$

With two unknowns one substitutes and eliminates by hand without much trouble; with three it is already fiddly and error-prone, and the systems that arise in practice—balancing a chemical equation, finding the currents in a circuit from Kirchhoff's laws, fitting a model to data—routinely have hundreds or thousands of unknowns. What we want is not a lucky trick but a *systematic, mechanical* procedure that always works and that a computer can carry out. That procedure is what this lecture builds. The whole system collapses into the single statement

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \qquad \mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix},$$

once we have said what the array $\mathbf{A}$ is and how it multiplies the column $\mathbf{x}$. We build that machinery now and return, in Section 3, to solve this system two different ways.

Remark (Matrices are how data is stored). Beyond solving systems, the same rectangular array is the everyday object of data science. A data set is a matrix $\mathbf{X}$ whose rows are samples and whose columns are features—each row a data point of the kind met in Lecture 4. A *transition matrix*, whose columns are probability distributions over states, drives the Markov-chain and PageRank models of Lecture 6. Fitting a straight line or a model to data leads to a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ solved by exactly the methods below. So the operations of this lecture are not only the engine of physics and engineering calculations but also the basic grammar of machine learning.

1 Matrices and their algebra

1.1 Definition and notation

Definition 1.1 (Matrix). An *$m \times n$ matrix* is a rectangular array of (real, and later complex) numbers with m rows and n columns,

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij}),$$

where a_{ij} is the *entry* in row i and column j. The pair $m \times n$ is its *size* (or *order*). Two matrices are *equal* when they have the same size and equal entries throughout.

A matrix with one column ($n = 1$) is a *column vector*, the objects of Lecture 4 written vertically; a matrix with one row is a *row vector*. When $m = n$ the matrix is *square* of order n, and its entries $a_{11}, a_{22}, \ldots, a_{nn}$ form the *main diagonal*.

1.2 Addition, subtraction and scalar multiplication

These are performed entrywise, exactly as for vectors, and are defined only for matrices of the *same* size.

Definition 1.2 (The entrywise operations). For $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ of the same size and a scalar k,

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij}), \qquad k\mathbf{A} = (k\,a_{ij}), \qquad \mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}.$$

Example 1.1 ***(Entrywise arithmetic).***

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 4 & -1 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 5 & 5 \end{pmatrix}, \qquad 3\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 0 & 9 \end{pmatrix}.$$

1.3 Matrix multiplication

Multiplication is the operation that makes matrices powerful, and it is *not* entrywise. The rule is built so that a linear system becomes $\mathbf{A}\mathbf{x} = \mathbf{b}$ (shown in the note below), and so that composing two transformations corresponds to multiplying their matrices—a payoff made precise once transformations are defined in §3.4.

Definition 1.3 (Matrix product). If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$—so that the number of *columns* of $\mathbf{A}$ equals the number of *rows* of $\mathbf{B}$—then the product $\mathbf{AB}$ is the $m \times p$ matrix whose (i, j) entry is the dot product of row i of $\mathbf{A}$ with column j of $\mathbf{B}$:

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Note (The compatibility rule). The product $\mathbf{AB}$ exists only when the inner sizes match, and the outer sizes give the result:

$$(\,m \times \underbrace{n)\;(n}_{\text{must agree}} \;\times p\,) \;\longrightarrow\; m \times p.$$

If the inner dimensions disagree, the product is undefined.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 10 & 11 \end{pmatrix}$$

$$(\mathbf{AB})_{11} = 1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 = 4$$

Figure 1: Each entry of **AB** is a row of **A** dotted with a column of **B**; the shaded row and column produce the shaded entry.

Example 1.2 ***(A compatible product).*** A 2×3 matrix times a 3×2 matrix gives a 2×2 matrix (Figure 1):

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1+0+3 & 0+2+3 \\ 4+0+6 & 0+5+6 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 10 & 11 \end{pmatrix}.$$

1.4 Properties, and one that fails

Matrix multiplication is associative and distributes over addition. Unlike ordinary multiplication, however, it is *not commutative*.

Theorem 1.1 (Algebraic properties of the product). Whenever the sizes are compatible,

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}), \qquad \mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB} + \mathbf{AC}, \qquad (\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}.$$

In general, however, $\mathbf{AB} \neq \mathbf{BA}$.

Example 1.3 ***(Order matters).*** With $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$,

$$\mathbf{AB} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}, \qquad \mathbf{BA} = \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix},$$

so $\mathbf{AB} \neq \mathbf{BA}$. The order of multiplication is part of the information.

Note (A system is a product). With multiplication defined, the motivating system is literally $\mathbf{Ax} = \mathbf{b}$: the ith entry of $\mathbf{Ax}$ is row i of $\mathbf{A}$ dotted with the column of unknowns, reproducing the ith equation. Storing a system as a matrix is not just shorthand—it lets the algebra of matrices do the solving.

1.5 Special matrices

Certain shapes recur often enough to be named.

Definition 1.4 (A vocabulary of matrices).

- The *zero matrix* $\mathbf{O}$ has every entry 0; it is the additive identity, $\mathbf{A} + \mathbf{O} = \mathbf{A}$.
- The *identity matrix* $\mathbf{I}_n$ is the square matrix with 1s on the main diagonal and 0s elsewhere; it is the multiplicative identity: for a square $\mathbf{A}$, $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$ (for a rectangular $\mathbf{A}$ the left and right identities differ in size).
- A *diagonal* matrix has zero off the main diagonal; a matrix is *upper (lower) triangular* if all entries below (above) the diagonal are zero.

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \text{ (diagonal)}, \quad \begin{pmatrix} 1 & 4 & 7 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{pmatrix} \text{ (upper triangular)}.$$

1.6 Transpose and trace

Definition 1.5 (Transpose). The *transpose* $\mathbf{A}^\mathsf{T}$ of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix obtained by interchanging rows and columns: the (i, j) entry of $\mathbf{A}^\mathsf{T}$ is a_{ji}. A square matrix with $\mathbf{A}^\mathsf{T} = \mathbf{A}$ is *symmetric*.

Example 1.4 ***(A transpose).***

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \; (2 \times 3) \qquad \Longrightarrow \qquad \mathbf{A}^\mathsf{T} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \; (3 \times 2).$$

The transpose passes through sums unchanged but *reverses the order* of a product—a small fact that will matter repeatedly.

Proposition 1.1 (Properties of the transpose).

$$(\mathbf{A}^\mathsf{T})^\mathsf{T} = \mathbf{A}, \qquad (\mathbf{A} + \mathbf{B})^\mathsf{T} = \mathbf{A}^\mathsf{T} + \mathbf{B}^\mathsf{T}, \qquad (k\mathbf{A})^\mathsf{T} = k\,\mathbf{A}^\mathsf{T}, \qquad (\mathbf{AB})^\mathsf{T} = \mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}.$$

Definition 1.6 (Trace). The *trace* of a square matrix is the sum of its diagonal entries, $\operatorname{tr}(\mathbf{A}) = a_{11} + a_{22} + \cdots + a_{nn}$.

The trace is linear, $\operatorname{tr}(\mathbf{A} + \mathbf{B}) = \operatorname{tr}(\mathbf{A}) + \operatorname{tr}(\mathbf{B})$, and has the useful cyclic property $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$ even though the products themselves differ. Both the trace and the determinant of the next section will reappear in Lecture 6 as quantities a matrix carries with it through any change of coordinates.

Example 1.5 ***(Trace and the cyclic property).*** For $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 4 & 1 \\ 1 & 0 \end{pmatrix}$, $\operatorname{tr}\mathbf{A} = 1 + 3 = 4$. The products differ, $\mathbf{AB} = \begin{pmatrix} 6 & 1 \\ 3 & 0 \end{pmatrix}$ and $\mathbf{BA} = \begin{pmatrix} 4 & 11 \\ 1 & 2 \end{pmatrix}$, yet $\operatorname{tr}(\mathbf{AB}) = 6 = \operatorname{tr}(\mathbf{BA})$, as the cyclic property promises.

2 The determinant and the inverse

To every *square* matrix we attach a single number, the determinant, that decides at a glance whether the matrix can be undone—whether it has an inverse, and hence whether $\mathbf{Ax} = \mathbf{b}$ has a unique solution.

2.1 The determinant

Definition 2.1 (Determinant of a 2×2 matrix).

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Remark (The determinant is the area and volume of Lecture 4). This is no coincidence with the cross product. The number $ad - bc$ is the signed area of the parallelogram spanned by the columns (a, c) and (b, d)—precisely the z-component of their cross product. In three dimensions the determinant of a matrix is the signed *volume* of the parallelepiped on its columns, the scalar triple product of Lecture 4. A determinant of zero therefore means a collapsed, flattened figure—and, as we are about to see, a matrix that cannot be inverted.

For larger matrices the determinant is defined by *cofactor expansion*, reducing an $n \times n$ determinant to smaller ones.

Definition 2.2 (Minors, cofactors and expansion). The *minor* M_{ij} of a square matrix is the determinant of the submatrix left when row i and column j are deleted, and the *cofactor* is the signed minor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

the signs following the checkerboard pattern $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$. The determinant is then defined by expansion along the *first* row,

$$\det \mathbf{A} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n},$$

the cofactors C_{1j} being determinants of smaller matrices, so the definition unwinds down to the 2×2 case already given.

Note (Expand along whichever row or column is easiest). It is a theorem (proved in the Linear Algebra module) that expanding along *any* row or *any* column gives the same number,

$$\det \mathbf{A} = a_{i1}C_{i1} + \cdots + a_{in}C_{in} = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj},$$

so in practice we pick the row or column with the most zeros. We use this freely below.

Example 2.1 ***(A 3×3 determinant).*** For the coefficient matrix of our system, expand along the first row:

$$\det \mathbf{A} = \begin{vmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{vmatrix} = 2\begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} - 1\begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} + (-1)\begin{vmatrix} -3 & -1 \\ -2 & 1 \end{vmatrix}.$$

The three 2×2 determinants are $(-2-2) = -4$, $(-6+4) = -2$ and $(-3-2) = -5$, so

$$\det \mathbf{A} = 2(-4) - 1(-2) + (-1)(-5) = -8 + 2 + 5 = -1.$$

Since $\det \mathbf{A} \neq 0$, the matrix is invertible and the system has a unique solution.

2.2 The inverse

Definition 2.3 (Inverse matrix). A square matrix $\mathbf{A}$ is *invertible* (or *non-singular*) if there is a matrix $\mathbf{A}^{-1}$ with

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Such an $\mathbf{A}^{-1}$ is unique when it exists, and exists precisely when $\det \mathbf{A} \neq 0$.

For a 2×2 matrix the inverse is immediate: swap the diagonal, negate the off-diagonal, and divide by the determinant.

Theorem 2.1 (Inverse of a 2×2 matrix). If $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \neq 0$, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Example 2.2 ***(A 2×2 inverse).*** For $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ the determinant is $6 - 5 = 1$, so

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix},$$

and indeed $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

For a 3×3 matrix the same idea works with cofactors: the inverse is the transposed matrix of cofactors—the *adjugate*—divided by the determinant.

Note (The cofactor method for $\mathbf{A}^{-1}$).

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}\,\mathrm{adj}(\mathbf{A}), \qquad \mathrm{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}},$$

where $\mathbf{C} = (C_{ij})$ is the matrix of cofactors. (For larger matrices the row-reduction method of Section 3, applied to $(\,\mathbf{A} \mid \mathbf{I}\,)$, is far quicker; the cofactor formula is the hand method for

3×3.)

Example 2.3 ***(A 3×3 inverse, with verification).*** Invert $\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}$, for which $\det \mathbf{A} = -1$.

Cofactors. Deleting each row and column and attaching the checkerboard sign gives the nine cofactors

$$\mathbf{C} = \begin{pmatrix} -4 & 2 & -5 \\ -3 & 2 & -4 \\ 1 & -1 & 1 \end{pmatrix},$$

for instance $C_{11} = +\begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} = -4$ and $C_{12} = -\begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} = +2$.

Adjugate and inverse. Transpose $\mathbf{C}$ and divide by $\det \mathbf{A} = -1$:

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}} = \begin{pmatrix} -4 & -3 & 1 \\ 2 & 2 & -1 \\ -5 & -4 & 1 \end{pmatrix}, \qquad \mathbf{A}^{-1} = \frac{1}{-1}\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix}.$$

Verification. Multiplying confirms the inverse:

$$\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}.$$

We keep $\mathbf{A}^{-1}$ for the next section, where it solves the system in one stroke.

3 Linear systems and transformations

We can now keep the promise of the opening. A linear system is $\mathbf{A}\mathbf{x} = \mathbf{b}$, and there are two routes to its solution: invert $\mathbf{A}$, or eliminate.

3.1 The inverse-matrix method

If $\mathbf{A}$ is invertible, multiplying $\mathbf{A}\mathbf{x} = \mathbf{b}$ on the left by $\mathbf{A}^{-1}$ isolates $\mathbf{x}$:

$$\mathbf{A}\mathbf{x} = \mathbf{b} \implies \mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \implies \mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Example 3.1 ***(The system by the inverse).*** With $\mathbf{A}^{-1}$ from Section 2 and $\mathbf{b} = (8, -11, -3)^{\mathsf{T}}$,

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix} \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix} = \begin{pmatrix} 32 - 33 + 3 \\ -16 + 22 - 3 \\ 40 - 44 + 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

So $x = 2$, $y = 3$, $z = -1$. The method is elegant, but it makes us compute a whole inverse to extract one solution vector. When the goal is simply to solve the system, elimination does less work.

3.2 Gaussian elimination

Elimination works directly on the *augmented matrix* $(\,\mathbf{A} \mid \mathbf{b}\,)$, simplifying it with operations that never change the solution set.

Definition 3.1 (Elementary row operations). Each of the following turns a system into an equivalent one:

1. swap two rows;
2. multiply a row by a nonzero scalar;
3. add a multiple of one row to another.

A matrix is in *row echelon form* when each leading (leftmost nonzero) entry sits to the right of the one above it and rows of zeros are at the foot; clearing the entries *above* each leading entry as well, and scaling each to 1, gives *reduced* row echelon form.

Gaussian elimination reduces $(\,\mathbf{A} \mid \mathbf{b}\,)$ to a triangular (echelon) form and then *back-substitutes*; carrying the reduction all the way to reduced echelon form (*Gauss–Jordan elimination*) lets the solution be read straight off.

Example 3.2 ***(The system by Gaussian elimination).*** Reduce the augmented matrix, using operations that keep the entries integers:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right) \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to 2R_2 + 3R_1} \left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 5 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 1 \end{array}\right).$$

The matrix is now triangular. The last row reads $-z = 1$, so $z = -1$; the second row gives $y + z = 2$, so $y = 3$; and the first gives $2x + y - z = 8$, that is $2x + 3 + 1 = 8$, so $x = 2$. The solution is

$$(x, y, z) = (2,\ 3,\ -1),$$

exactly the inverse-matrix answer—two roads to the same destination, and this one needed no inverse. (Gauss–Jordan would continue, clearing the upper entries and scaling, to reach $(\,\mathbf{I} \mid \mathbf{x}\,)$ with the solution displayed directly.)

Note (Row operations and the determinant). The three elementary operations have predictable effects on the determinant, which is part of why elimination can be trusted: swapping two rows multiplies det by -1; multiplying a row by k multiplies det by k; and adding a multiple of one row to another leaves det unchanged. The reduction just performed illustrates all three. Only the step $R_2 \to 2R_2 + 3R_1$ altered the size of the determinant—it scaled row 2 by 2, the added multiple of row 1 contributing nothing—while every other operation (each an add-a-multiple) left it fixed. The matrix produced is triangular, with determinant the diagonal product $2 \cdot 1 \cdot (-1) = -2$; this equals $2 \det \mathbf{A}$, so $\det \mathbf{A} = -1$, agreeing with the cofactor expansion of Section 2.

The same row reduction also *inverts* a matrix, delivering on the promise of the note in Section 2.2. To find $\mathbf{A}^{-1}$, reduce the augmented matrix $(\,\mathbf{A} \mid \mathbf{I}\,)$ all the way to $(\,\mathbf{I} \mid \mathbf{A}^{-1}\,)$: the very operations that drive $\mathbf{A}$ to the identity on the left carry the identity to $\mathbf{A}^{-1}$ on the right.

Example 3.3 ***(The inverse by row reduction).*** We invert the running matrix a third way, using the same opening operations as the elimination above. First reduce the left block to upper-triangular form:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ -3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to 2R_2 + 3R_1} \left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 0 & -1 & -5 & -4 & 1 \end{array}\right).$$

Now continue (Gauss–Jordan) by scaling row 3 and clearing the third column above it:

$$\xrightarrow{R_3 \to -R_3} \left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array}\right) \xrightarrow[R_1 \to R_1 + R_3]{R_2 \to R_2 - R_3} \left(\begin{array}{ccc|ccc} 2 & 1 & 0 & 6 & 4 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array}\right).$$

Finally clear the second column of row 1 and scale row 1 to a leading 1:

$$\xrightarrow{R_1 \to R_1 - R_2} \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 8 & 6 & -2 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array}\right) \xrightarrow{R_1 \to \frac{1}{2} R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 3 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array}\right).$$

The left block is now $\mathbf{I}$, so the right block is the inverse,

$$\mathbf{A}^{-1} = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix},$$

exactly the matrix found by cofactors in Section 2—a third road to the same destination, and the one that scales to large systems where cofactors become hopeless.

3.3 What the solutions look like

A single linear equation in two unknowns is a line, and in three unknowns a plane (Lecture 4). Solving a system asks where these meet, and there are exactly three possibilities, visible already for two lines in the plane (Figure 2).

Note (Reading the echelon form). Elimination tells the three cases apart automatically. A full set of leading entries (one per unknown, as in our example) gives a *unique* solution. A row $(\,0\;0\;\cdots\;0 \mid c\,)$ with $c \neq 0$ reads $0 = c$, an impossibility: *no* solution. A row of zeros $(\,0\;0\;\cdots\;0 \mid 0\,)$ leaves an unknown free to take any value: *infinitely many* solutions. The determinant flags the first case in advance: $\det \mathbf{A} \neq 0$ guarantees a unique solution.

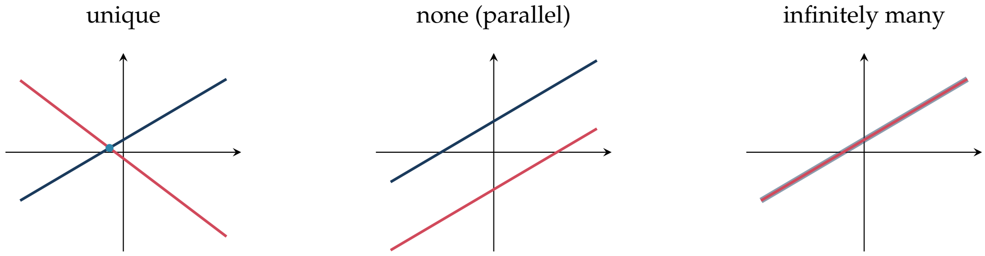

Figure 2: Two lines meet in one point (unique solution), never (no solution, parallel), or everywhere (infinitely many, the same line). Three planes give the same three *counts*—though for planes the no-solution case can also occur with no two of them parallel (they bound a triangular prism).

Example 3.4 ***(A system with infinitely many solutions).*** Not every consistent system pins down a single point. Consider

$$\begin{aligned} x + y + z &= 6, \\ x + 2y + 3z &= 14, \\ 2x + 3y + 4z &= 20. \end{aligned}$$

Reducing the augmented matrix with the same elimination as before,

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 20 \end{array}\right) \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & 2 & 8 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The last row has become all zeros: the third equation was the sum of the first two and carried no new information, so one unknown is left *free*. Set $z = t$. The second row then gives $y + 2z = 8$, so $y = 8 - 2t$, and the first gives $x = 6 - y - z = -2 + t$. The solution is a whole one-parameter family,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R},$$

which is exactly the vector equation of a *line* from Lecture 4: the three planes meet not in a point but along a common line. (Put $t = 0, 1, 2$ to read off the points $(-2, 8, 0)$, $(-1, 6, 1)$, $(0, 4, 2)$, each of which satisfies all three equations.) Had the right-hand side instead produced a row $\left(\begin{array}{ccc|c} 0 & 0 & 0 & c \end{array}\right)$ with $c \neq 0$, the very same elimination would have signalled *no* solution.

Remark (Looking ahead: rank and the general theory). The number of leading entries left after elimination—the count of genuinely independent equations—is called the *rank* of the matrix, and it is what decides between the three cases for a system of any size. We use it here only informally, through the shape of the echelon form. The systematic theory of rank, of general $m \times n$ systems, and of abstract vector spaces (independence, basis and dimension, first met geometrically in Lecture 4) is developed in the dedicated *Linear Algebra module*, for which this lecture is the computational foundation.

3.4 Matrices as transformations

A square matrix does more than store a system: multiplying a vector by an $n \times n$ matrix sends $\mathbb{R}^n$ to itself, and the map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ is a *linear transformation*—it preserves vector addition and scaling. In the plane, the 2×2 matrix is read off by seeing where it sends the basis vectors $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$: those images are its columns.

Note (The standard plane transformations).

$$\underbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}_{\text{rotation by } \theta} \quad \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\text{reflect in } y=x} \quad \underbrace{\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}}_{\text{enlargement } \times k} \quad \underbrace{\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}}_{\text{stretch}} \quad \underbrace{\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}}_{\text{shear}}$$

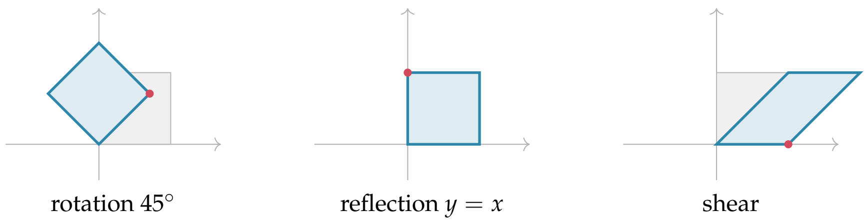

Figure 3: The unit square (grey) under three transformations; the marked corner is the image of $\mathbf{i}$, the first column of the matrix.

Example 3.5 ***(Composing and reading a transformation).*** A rotation by 90° has matrix $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and it sends $\mathbf{i} = (1, 0)$ to $(0, 1)$ and $\mathbf{j} = (0, 1)$ to $(-1, 0)$—a quarter-turn, the real-plane echo of multiplying by i in Lecture 3. Applying one transformation after another corresponds to *multiplying* the matrices, in right-to-left order: doing $\mathbf{S}$ then $\mathbf{R}$ is the single matrix $\mathbf{RS}$. Because the determinant scales area, $\det \mathbf{R} = 1$ keeps area and orientation, a reflection has $\det = -1$ (area kept, orientation flipped), and a singular matrix ($\det = 0$) collapses the plane onto a line.

Remark (Translations are not linear). A translation $\mathbf{x} \mapsto \mathbf{x} + \mathbf{t}$ shifts every point by a fixed vector, but it is *not* a linear transformation: it moves the origin, whereas $\mathbf{A0} = \mathbf{0}$ always. Such maps are called *affine*. The standard fix—used throughout computer graphics—is to lift the plane into $\mathbb{R}^3$ with an extra coordinate 1, after which a single 3×3 matrix can carry out rotation, scaling *and* translation together; but within $\mathbb{R}^2$, translation lies outside the reach of a 2×2 matrix.

Summary

Concept	Key idea
Matrix	$m \times n$ array (a_{ij}); entrywise $+$ and scalar $\times$ (same size).
Multiplication	$(\mathbf{AB})_{ij}$ = row i of $\mathbf{A}$ $\cdot$ column j of $\mathbf{B}$; needs columns of $\mathbf{A}$ = rows of $\mathbf{B}$; $(m\times n)(n\times p) \to m\times p$.
Properties	Associative and distributive, but *not* commutative: $\mathbf{AB} \neq \mathbf{BA}$ in general.
Special	Zero $\mathbf{O}$, identity $\mathbf{I}$ ($\mathbf{AI} = \mathbf{A}$), diagonal, triangular.
Transpose	Swap rows/columns; $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$; symmetric if $\mathbf{A}^\top = \mathbf{A}$. Trace $\operatorname{tr}\mathbf{A} = \sum a_{ii}$.
Determinant	2×2: $ad - bc$ (signed area). Larger: cofactor expansion $\det\mathbf{A} = \sum_j a_{ij}C_{ij}$, $C_{ij} = (-1)^{i+j}M_{ij}$.
Inverse	Exists iff $\det\mathbf{A} \neq 0$; 2×2 formula $\frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$; 3×3 via $\mathbf{A}^{-1} = \frac{1}{\det\mathbf{A}}\operatorname{adj}\mathbf{A}$.
System, two ways	$\mathbf{Ax} = \mathbf{b}$: inverse method $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, or Gaussian elimination on $(\mathbf{A} \mid \mathbf{b})$ then back-substitute.
Solution types	Unique ($\det \neq 0$), none (a $0 = c$ row), or infinitely many (a free variable)—meeting in a point, not at all, or along a line/plane.
Transformations	$\mathbf{x} \mapsto \mathbf{Ax}$ is linear; columns are the images of $\mathbf{i}, \mathbf{j}$; rotation/reflection/stretch/shear; composition = product; $\lvert\det\rvert$ = area scale. Translation is affine, not linear.

Next lecture: eigensystems—the special directions a matrix merely stretches. Using the determinant for the characteristic equation, elimination for the eigenvectors, and the inverse for diagonalisation, we find the axes along which $\mathbf{Ax} = \lambda\mathbf{x}$*, and use them to compute matrix powers and the long-run behaviour of evolving systems.*

Companion material: a separate *Exercise Sheet 5* (with solutions) develops the practice problems for this unit.